A Multi-Fidelity Simulation Framework for X-Band Pulse Compression Radar

J.C. Vaught*, Douglas Cahl†, Yi Wang‡
Department of Mechanical Engineering, University of South Carolina, Columbia, SC, 29208

As surface vessels operate in dynamic and cluttered environments, the demand for robust radar-based perception has outpaced the availability of labeled training data. Collecting diverse radar datasets with reliably accurate ground truth is logistically complex, expensive, and constrained by safety considerations, yet modern detection and tracking algorithms require vast amounts of such data to generalize across sea states, object types, and clutter regimes. We present a multi-fidelity simulation framework for X-band radar designed to bridge this gap. The framework comprises three computational tiers: (i) a ray-tracing engine that simulates the electromagnetic pulse interactions with 3D scene geometry including material-dependent scattering and multipath effects; (ii) an analytical generator using statistical clutter models and stochastic target cross-section representations for rapid, large-scale data generation; and (iii) a data-driven compositor that blends real object signatures into existing radar recordings, preserving real sensor noise and texture. Each tier addresses a different stage of the algorithm development cycle, namely physics-grounded validation, high-throughput training data generation, and domain-transfer verification. We describe the architecture and computational trade-offs of each mode and demonstrate their use for benchmarking detection and tracking methods for small maritime objects across representative sea-clutter and multipath conditions. The framework supports physics-grounded validation, high-throughput synthetic data generation, and domain-transfer checks using recorded radar data, enabling controlled studies of clutter rejection, object identification, and intermittent-return tracking through configurable scene geometry, sensor parameters, and fidelity settings.

*Graduate Student, Department of Mechanical Engineering, AIAA Student Member, Member ID: 1537189.
†Professor, Department of Mechanical Engineering.
‡Professor, Department of Mechanical Engineering.

I. Introduction

II. Related Work

III. System Architecture and World Model

IV. Tier 1: High-Fidelity Electromagnetic Ray Tracing

V. Tier 2: Analytical Approximation

VI. Tier 3: Data-Driven Compositor

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X. Conclusion and Future Work

Acknowledgments